

STAT 4102 Midterm 2

2020 Fall

1 (25 points)

Let $X_1, \dots, X_n$ be i.i.d. random variables with pdf

$$f(x) = \theta x^{\theta-1}, x \in (0, 1), \theta \in (0, \infty).$$

1. What is the likelihood function? (4 points)
2. Find a one-dimensional sufficient statistic for θ . Note that the vector $(X_1, \dots, X_n)$ is sufficient by definition, but it is n -dimensional. (7 points)
3. Find the method of moment estimator for θ . (7 points)
4. It is consistent? Prove your result. (7 points)

Solution:

1. $L(\theta|x_1, \dots, x_n) = \theta^n (\prod_i x_i)^{\theta-1}$.
2. By the factorization theorem, $\prod_i X_i$ is a sufficient statistic.
3. $EX_1 = \frac{\theta}{\theta+1}$, and thus a method of moments estimator is

$$\hat{\theta} = \frac{\bar{X}}{1 - \bar{X}}.$$

4. It is consistent. By the law of large number, $\bar{X} \rightarrow_p EX_1$. Then by theorem 9.2, we get $\frac{\bar{X}}{1-\bar{X}} \rightarrow_p \theta$, since $f(x) = \frac{x}{1-x}$ is continuous on $(0, 1)$.

2 (25 points)

Let $X_1, \dots, X_n$ be i.i.d. random variables with pdf

$$f(x) = \frac{2x}{\theta^2}, x \in (0, \theta), \theta \in (0, \infty).$$

1. What is the likelihood function? (4 points)
2. Find the MLE for θ . (10 points)
3. Is it consistent? Prove your result. (6 points)
4. Find the MLE for $\exp(\theta^2)$. Is it consistent? (5 points)

Solution:

1. $L(\theta|x_1, \dots, x_n) = \frac{2^n}{\theta^{2n}} \prod_i x_i 1_{x_i \leq \theta} 1_{x_i > 0} = \frac{2^n}{\theta^{2n}} 1_{\min x_i > 0} 1_{\max x_i \leq \theta} \prod_i x_i$.
2. $L(\theta) = (\prod_i x_i 1_{\min x_i > 0} 2^n) (1_{\max x_i \leq \theta} \frac{1}{\theta^{2n}})$. Since the first factor of the right hand side is a positive constant, the second factor is a decreasing function on $[\max x_i, \infty)$, and it is 0 when $\theta < \max x_i$, we know the likelihood function is maximized at $\max x_i$. Thus $\hat{\theta} = \max X_i$.
3. Let $\epsilon > 0$. Because $\max X_i \leq \theta$, we have

$$P(|\max X_i - \theta| > \epsilon) = P(\max X_i < \theta - \epsilon).$$

Now, let $F(x)$ be the c.d.f. of X_1 . We know that $F(x) < 1$, for any $x < \theta$. Since

$$P(\max X_i < \theta - \epsilon) = P(X_1, X_2, \dots, X_n < \theta - \epsilon) = P(X_1 \leq \theta - \epsilon)^n = F(\theta - \epsilon)^n,$$

we get

$$P(|\max X_i - \theta| > \epsilon) \rightarrow 0, \text{ as } n \rightarrow \infty.$$

4. By the invariance property of MLE, the MLE of $\exp(\theta^2)$ is $\exp\{(\max X_i)^2\}$. We've shown that $\max X_i \rightarrow_p \theta$. By theorem 9.2, the MLE is again consistent.

3 (25 points)

Let $X_1, \dots, X_n$ be i.i.d. Gamma(α, β) distributed and $\alpha > 0$ is known. The p.d.f. is

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta), x > 0.$$

1. Find the MLE of β . (5 points)
2. Suppose we want to test $H_0 : \beta = \beta_0$ against $H_1 : \beta \neq \beta_0$, for some known and fixed $\beta_0 > 0$. Develop a level α test for it, using the MLE as the test statistic. (15 points)
3. What is the power of your test at the point $\beta_1 \neq \beta_0$? (5 points)

You may use the fact that $\sum_i^n X_i \sim \text{Gamma}(n\alpha, \beta)$.

Solution:

1. The log-likelihood function is

$$l(\beta|x_1, \dots, x_n) = -n \log(\Gamma(\alpha)) - n\alpha \log(\beta) + (\alpha - 1) \left(\sum_i \log(x_i) \right) - \frac{1}{\beta} \sum_i x_i.$$

Taking a derivative w.r.t. β , we have

$$-n\alpha \frac{1}{\beta} + \frac{1}{\beta^2} \sum_i x_i.$$

By setting it to 0, we get $\frac{\sum X_i}{n\alpha}$. Because the second derivative

$$n\alpha \frac{1}{\beta^2} - \frac{2}{\beta^3} \sum_i x_i$$

is negative at this point, we know that the MLE is

$$\hat{\beta} = \frac{\sum X_i}{n\alpha}.$$

2. Let $\gamma_{1-\omega/2}$ and $\gamma_{\omega/2}$ be two quantiles of the $\text{Gamma}(n\alpha, \beta_0)$ distribution. Then we have

$$P_{H_0}(\gamma_{1-\omega/2}/(n\alpha) \leq \hat{\beta} \leq \gamma_{\omega/2}/(n\alpha)) = \omega.$$

Thus we can reject the null when $\hat{\beta} \leq \gamma_{1-\omega/2}/(n\alpha)$ or $\hat{\beta} \geq \gamma_{\omega/2}/(n\alpha)$, and this is a level ω test.

3. The power is

$$P(\text{Gamma}(n\alpha, \beta_1) \geq \gamma_{\omega/2}) + P(\text{Gamma}(n\alpha, \beta_1) \leq \gamma_{1-\omega/2}).$$

4 (25 points)

A coin operated soda machine was designed to discharge an average of 7 ounces of soda per cup. In a test of the machine, 30 cups of beverage were drawn from the machine and measured. The mean and standard deviation of the measurements were 7.1 ounces and 0.12 ounce, respectively.

1. Do these data present sufficient evidence to indicate that the mean discharge differs from 7 ounces? Use $\alpha = 0.10$. If you use a formula for this problem, please verify every condition needed for it. (20 points)
2. What is the p-value? Write your answer in terms of the c.d.f of normal distribution. You do not need to calculate the number. (5 points)

(Some quantiles you may need: $z_{0.25} = 1.96$, $z_{0.05} = 1.64$, $z_{0.1} = 1.28$)

Solution:

1. Let $\bar{X}$ be the sample mean, and S^2 be the sample variance. We use the statistic

$$Z = \frac{\bar{X} - 7}{\sqrt{S}/\sqrt{n}}$$

for testing $H_0 : \mu = 7$ vs $H_1 : \mu \neq 7$. If we assume that the volume of beverage in the 30 cups $X_1, \dots, X_n$ are i.i.d., with mean μ and variance σ^2 , then under the null hypothesis, by the central limit theorem,

$$\bar{X} - \mu \sim_{approx} N(0, \sigma^2/n).$$

Since S^2 is a consistent estimator for σ^2 , we have that under the null hypothesis,

$$Z = \frac{\bar{X} - 7}{\sqrt{S}/\sqrt{n}} \sim_{approx} N(0, 1).$$

Let z_α denote the corresponding quantile of the standard normal distribution. Then we know a test that rejects the null when $|Z| \geq z_{\alpha/2}$ is a level α test. Using the data provided, we have $Z = 4.564$ and $z_{0.05} = 1.64$. Thus we reject the null hypothesis.

2. Let ϕ be the c.d.f. of the standard normal distribution. The p-value is $2\phi(-|Z|)$.